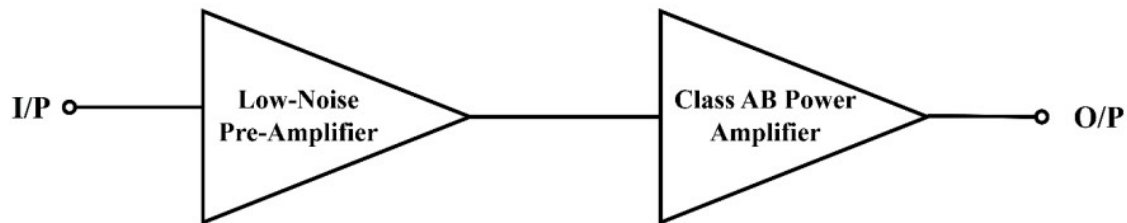


EE-313 Electronic Circuit Design: Project Statement

Design an audio amplifier which will take input from a small microphone and can drive a small speaker. A condenser microphone can be used. The audio amplifier must be based on discrete devices and if needed audio transformers may be used. However, op-amp or IC amplifiers CANNOT be used. Moreover, base your design on locally available BJTs and MOSFETs. Justify selection of transistors for each specific stage.

Design Specifications:

1. AC output power = 1 to 3 Watt (8 Ω speaker)
2. Frequency response of the amplifier should be 100 Hz to 12 KHz at 3 dB level.
3. The audio amplifier should qualify the given experimental testing standard.



DC Power Supply: $\pm 18V$

Overall Amplifier Gain: 500 V/V

Pre-Amplifier Specifications:

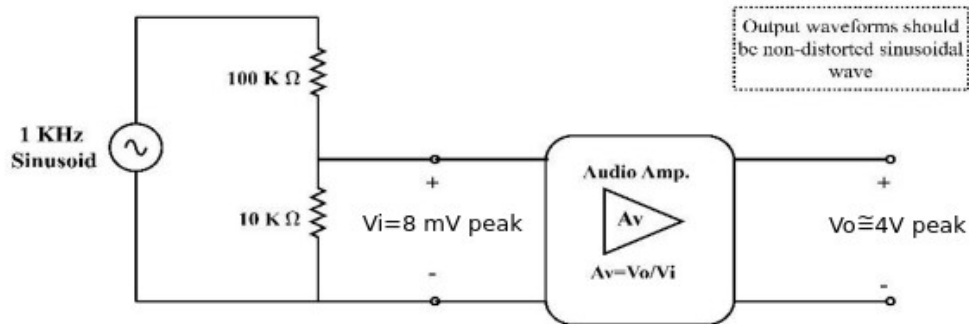
- $R_{in} \geq 100K\Omega$, $R_o \leq 5K\Omega$

Power Stage:

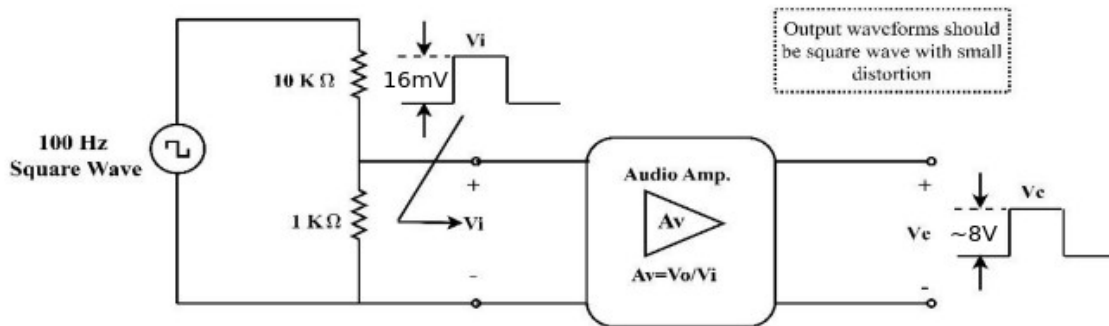
- $R_L = 8\Omega$
- V_o (signal swing) = $\pm 4V$

Audio amplifier experimental testing standard

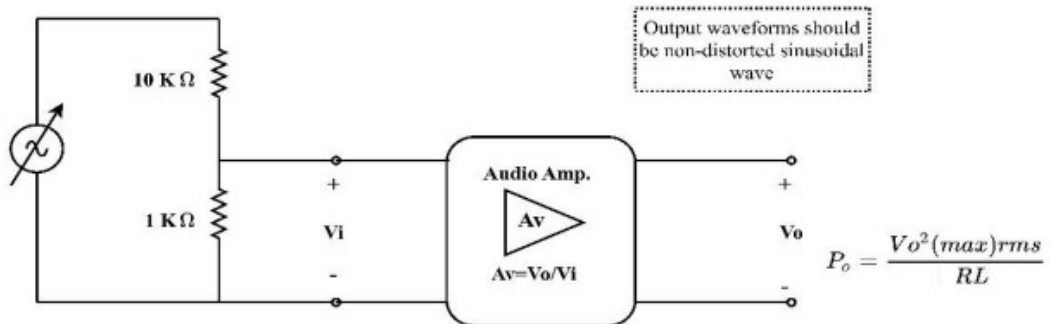
(A) A_v Test



(B) Frequency Response Test



(C) Maximal Output Power, $P_o(\text{max})$



Note: Increase input voltage magnitude until reaching the maximum output voltage $V_{o(\text{max})}$, which is the non distorted sinusoidal wave.